

SANJAI K

Chennai, India, 600052 | sanjaikumarmca@gmail.com | +91 9789544568

GitHub: <https://github.com/sanjaikmca> | LinkedIn: <https://www.linkedin.com/in/sanjai-k-9159a02a8/>

PROFESSIONAL SUMMARY

Passionate about transforming data into meaningful insights, I am an MCA graduate with hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing. Proficient in Python, TensorFlow, Scikit-learn, and data analysis, with practical experience developing and evaluating machine learning models on real-world datasets. Presented research work at an IEEE International Conference and eager to apply my technical skills and problem-solving abilities to real-world challenges.

INTERNSHIPS & EXPERIENCE

Research Intern – Artificial Intelligence

Aug – Oct 2025

Hindustan Institute of Technology & Science, Chennai

- ✓ Built a Chest X-ray classification model using MobileNet, TensorFlow, and Grad-CAM for medical image analysis.

Data Science Intern

July 2025

CodeAlpha (Remote)

- ✓ Developed ML mini-projects using Python, Pandas, and Scikit-learn involving prediction, classification, and data analysis tasks.

KEY SKILLS

Programming Languages : Python, MySQL, HTML, CSS

Machine Learning & AI : Supervised Learning, Unsupervised Learning, Classification, Regression, Feature Engineering, Model Evaluation

Deep Learning & Computer Vision : CNN, LSTM, U-Net, Transfer Learning, Image Segmentation, Object Detection, Computer Vision

Natural Language Processing (NLP) : Text Preprocessing, Tokenization, Text Vectorization, Sentiment Analysis, Text Classification

Frameworks & Libraries : TensorFlow, Keras, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn

Tools & Platforms : Jupyter Notebook, Google Colab, Git, GitHub

Professional Skills : Analytical Thinking, Leadership, Team Collaboration, Research Documentation, Time Management, Adaptability, Quick Learning, Communication

EDUCATION

Master of Computer Applications (MCA) - CGPA: 9.3

Aug 2024 – May 2026

Hindustan Institute of Technology & Science, Chennai

Bachelor of Computer Applications (BCA) - CGPA: 7.9

Aug 2021 – May 2024

Dr. M.G.R Chockalingam Arts College, Arni

PROJECTS & RESEARCH

Brain Tumor Segmentation in MRI Scans | Deep U-Net (IEEE Published)

May 2026

ICCRTEE 2026 – IEEE Madras Section & IEEE Robotics and Automation Society

- ✓ Designed and trained a U-Net deep learning model on the Kaggle LGG MRI dataset (3,000+ real-world medical samples).
- ✓ Built a custom OpenCV preprocessing pipeline with data augmentation; optimized using Dice Loss + Adam optimizer.

X-Ray Image Classification | MobileNet Transfer Learning (Medical AI)

Oct 2025

Computer Vision – Healthcare Domain

- ✓ Developed medical image classifier using MobileNet transfer learning with MD.ai DICOM API for real-world X-ray data integration.
- ✓ Achieved 85–90% validation accuracy using early stopping, learning rate scheduling, and fine-tuned hyperparameters.

Image Caption Generator | VGG16 + LSTM (Computer Vision + NLP)

Apr 2025

Deep Learning – Multimodal AI

- ✓ Built end-to-end multimodal pipeline combining VGG16 CNN image embeddings with LSTM-based sequence prediction.
- ✓ Trained on ~8,000 real-world image-caption pairs (MSCOCO); achieved BLEU score ~0.4 using a custom Keras data generator.

CERTIFICATIONS AND WORKSHOPS

Introduction to Algorithms and Analysis – Elite Certificate

NPTEL / IIT Kharagpur (SWAYAM)

Intellithon'25 – 24-Hour National Hackathon

Hindustan Institute of Technology & Science (Dept. of Intelligent Systems & Cybersecurity)

Cloud Partner: Zoho • 24-hour national-level event

Workshop: Practical Approaches to Data Analytics Using SQL and Cloud Services

Dept. of Computer Applications, Hindustan Institute of Technology & Science

ACHIEVEMENTS

- ✓ Academic Rank Holder - MCA, Hindustan Institute of Technology & Science, Chennai (CGPA: 9.3 / 10).
- ✓ IEEE Conference Presenter - ICCRTEE 2026, Kalasalingam Academy of Research and Education, Tamil Nadu.
- ✓ IEEE Published Researcher - Brain Tumor Segmentation paper under IEEE Madras Section & IEEE Robotics and Automation Society.